

GSM A5/1 Stream Cipher Implementation and Known-Plaintext Attack Simulator

Anis Boulfoul Houssem Beniakhlef Khaled Guessaoui

Wireless and Mobile Network Security

May 2, 2026

Project Goal

- Implement the GSM A5/1 stream cipher in Python.
- Simulate encryption/decryption of GSM-like frames.
- Demonstrate a known-plaintext attack in an educational reduced keyspace setup.
- Deliver a reliable live demo for presentation day.

A5/1 Cipher Design

- Three LFSRs:
 - R1: 19 bits
 - R2: 22 bits
 - R3: 23 bits
- Majority clocking controls irregular register updates.
- Initialization inputs:
 - 64-bit session key
 - 22-bit frame number
- Output keystream is XORed with plaintext/ciphertext.

What We Implemented

- `src/a51.py`: cipher core (LFSRs, clocking, keystream, encrypt/decrypt).
- `src/attack.py`: known-plaintext attack simulator.
- `src/demo.py`: one-command guided end-to-end demo.
- `src/cli.py`: command-line interface:
 - keystream, encrypt, decrypt
 - attack, attack-sweep, attack-rate, demo

Known-Plaintext Attack (Educational)

- Attacker knows a plaintext segment and matching ciphertext.
- Keystream segment is derived using XOR.
- Simulator brute-forces only unknown lower key bits (reduced keyspace).
- Returns candidate keys and attack runtime.
- Important: this is a pedagogical simulator, not a full real-world A5/1 break.

Reproducible Demo Scenario

- Real key: 0x123456789ABCDEF0
- Frame number: 0x13456
- Known plaintext: GSM-A51-DEMO-BLOCK
- Unknown bits for attack: 12 (4096 key candidates)
- Generated ciphertext:
6b82759166cc706f9c78326bc4fc140bbe18
- Recovered candidate key:
0x123456789ABCDEF0

Demo Flow (Live)

- 1 Show initial R1, R2, R3 register states.
- 2 Generate and show keystream.
- 3 Encrypt plaintext and show ciphertext (hex).
- 4 Run known-plaintext attack and recover key candidate.
- 5 Decrypt ciphertext and recover original message.

Main live command:

```
python3 -m src.cli demo
```

Evaluation Results

- Unit/integration tests passing: **7/7**.
- Attack sweep shows fewer candidates with longer known plaintext.
- In our fixed scenario:
 - 1 byte known plaintext: multiple candidates
 - 2+ bytes: unique key recovery in tested runs
- Attack-rate mode provides reproducible percentages over multiple trials.

Team Contribution

- **Anis Boulfoul:** A5/1 implementation and integration.
- **Houssem Beniakhlef:** attack simulation logic and evaluation.
- **Khaled Guessaoui:** demo workflow and presentation preparation.

Conclusion

- We built a complete, reproducible GSM A5/1 educational security demo.
- The project links cipher internals with an attack perspective.
- The live terminal demo is stable and presentation-ready.

Thank you.